Prabhavit Gupta

Applied AI Engineer: Autonomous Agents, Marketing Automation & Production LLMs



+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (CET Time Zone / EU Permanent Resident)

Summary

Applied AI Engineer with 3 years of hands-on production experience architecting autonomous AI agents, enterprise marketing automation tools, and tool-augmented LLM systems. Proven track record identifying operational bottlenecks across marketing, sales, and catalog workflows, building zero-to-one AI pipelines that automate repetitive tasks, synchronize third-party advertising APIs, and generate structured campaign assets. Deep practical expertise in Python, TypeScript, modern LLM function calling, vector retrieval pipelines, and end-to-end observability. Based in Venice, Italy (CET time zone) with an EU Long-Term Residence Permit (permanent unrestricted EU work authorization, no sponsorship required); available immediately for full-time remote roles.

Technical Expertise

- **Applied AI & Agent Automations:** Tool/function calling, autonomous agent workflows, LangChain, multi-turn reasoning chains, structured JSON outputs, prompt engineering, human-in-the-loop approval gates.
- **Marketing Systems & API Integration:** RESTful APIs, webhook consumers, advertising and marketing data connectors, third-party platform integrations, automated asset generation and validation pipelines.
- **Software Development & Backend Systems:** Python (FastAPI, asyncio, pandas, NumPy), TypeScript, Node.js, PostgreSQL schema modeling, Redis asynchronous task queues, JSON schema enforcement.
- **Data Pipelines & Retrieval:** Retrieval-Augmented Generation (RAG), vector embeddings, semantic search, metadata filtering, dataset cleaning, deduplication, and automated anomaly detection.
- **DevOps & Quality Engineering:** Docker containerization, CI/CD pipelines, Git version control, automated regression and contract testing with Playwright, Linux server management.
- **Languages & Authorization:** English (C1 Professional, fluent), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (unrestricted EU work authorization across all EU member states, no sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer & AI Systems Architect *London, UK (remote)*

- Designed and operated production customer-facing AI agents and data extraction pipelines for a marketplace platform serving 4,500+ active enterprise users, integrating platform catalogs with external APIs.
- Built automated document parsing and ingestion pipelines using LLMs, converting unstructured partner data and marketing collateral into validated JSON schemas.
- Engineered multi-step agent reasoning chains with strict tool execution constraints, schema validation, and human-in-the-loop approval checkpoints, ensuring zero ungrounded actions in production flows.
- Implemented automated evaluation harnesses evaluating system accuracy, grounding, and recall against production transcripts, systematically lowering edge-case error rates.
- Architected and maintained resilient TypeScript/Node.js backend services and PostgreSQL data layers, utilizing asynchronous Redis queues to handle high-throughput query spikes.
- Developed automated integration and end-to-end regression test suites with Playwright, verifying API contracts and preventing system regressions across continuous deployment cycles.

Urlink **Apr 2025 - Jul 2025**
Project Manager & Solutions Developer (Freelance) *Barcelona, Spain (remote)*

- Led technical discovery, solution architecture, and client-facing delivery of customized web solutions and automated data processing pipelines for European client platforms.
- Built automated data ingestion and transformation services connecting third-party APIs to PostgreSQL databases, maintaining high data freshness and integrity.
- Delivered interactive functional prototypes to validate user flows and technical requirements prior to sprint execution, accelerating cycle times.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Automation & Integration Developer *Venice, Italy*

- Built and deployed custom LLM integrations, marketing automation workflows, and webhook pipelines for European clients using Python, TypeScript, and n8n.
- Replaced brittle manual copy and asset workflows with automated LLM-assisted generation and extraction pipelines featuring comprehensive error handling and fallback routing.
- Authored technical architecture specifications, API documentation, and deployment guides to support client engineering teams in operating self-hosted workflows.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous training in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.